Type MS · Describe labour and the advantages of breastfeeding

Answer 1 of 5 · Question 1 part b.i · 3 marks

2025 JUN · Paper 02 · find part b.i in this mark scheme extract

1 (a)		• urethra; • testes; • uterus; • seminal vesicle; • vagina; • oviduct; • ovary;		7
(b)	(i)	• oxytocin causes uterus muscles to contract; • during childbirth/labour; • causes/stimulates lactation/release of milk; • emotional bonding;	Accept produced during sexual intercourse for mp4	3
	(ii)	• LH triggers ovulation/release of egg; maintains corpus luteum/ involved in production of progesterone;		2
				Total = 12

Question number	Answer	Notes	Marks

Type MS · Describe labour and the advantages of breastfeeding

Answer 2 of 5 · Question 2 part c · 1 mark

2020 JAN · Paper 01 · find part c in this mark scheme extract

number				
2 (a)	(i)	**The only correct answer is A** *B is incorrect because it is the cervix* *C is incorrect because it is the oviduct* *D is incorrect because it is the uterus wall*		1
	(ii)	**The only correct answer is D** *A is incorrect because it is the vagina* *B is incorrect because it is the cervix* *C is incorrect because it is the oviduct*		1
	(iii)	• oestrogen (1) • lining (1) • ovulation (1)		3
(b)		needed for the growth of bones (1)		1
(c)		ref antibodies (passed to baby)/contains the correct balance of nutrients/reduced risk of diabetes/allergies/obesity later in life/reference to bonding between mother and baby (1)		1
(d)	(i)	more breastfed babies in 2010 compared with 1940 (1)	Allow reverse argument A 78% vs 42% difference of 36%	1
	(ii)	• life expectancy shows a constant increase (1) • despite number of babies breastfed increasing and decreasing (1)		2

Total for Question 2 = 10 marks

Type MS · Describe labour and the advantages of breastfeeding

Answer 3 of 5 · Question 6 part g · 1 mark

2023 JUN · Paper 02 (regional R) · find part g in this mark scheme extract

6 (a)	• fertilisation; • egg and sperm fuse/gametes/join;		2
(b)	Any four from • mitosis; • DNA is replicated; • chromosomes move to the centre of the cell; • chromosomes are separated to opposite poles; • cell divides;	Allow one mark for prophase, metaphase, anaphase, telophase in the correct order Allow zygote/embyro divides	4
(c)	**D;** (the cells are able to differentiate into any type of body cell)		1
(d)	• (corpus luteum) produces progesterone; • to maintain uterus lining;		2
(e)	Any two from: • umbilical cord established; • transports nutrients/named nutrients/oxygen (to fetus/embryo); • removes waste products/named waste product (from fetus/embryo);		2
(f)	development of male sexual characteristics/genitalia;		1
(g)	**C;** (release of oxytocin from the pituitary gland)		1

Total for Question 6 = 13 marks

Type MS · Describe labour and the advantages of breastfeeding

Answer 4 of 5 · Question 6 part d · 2 marks

2024 JUN · Paper 01 (regional R) · find part d in this mark scheme extract

Question number			
6	(a)	<table><tr><th rowspan="2">Hormone</th><th colspan="4">Function of hormone</th></tr><tr><th>causes ovulation</th><th>builds up uterus lining after menstruation</th><th>maintains uterus lining during pregnancy</th><th>stimulates a follicle to mature</th></tr><tr><td>progesterone</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>FSH</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>oestrogen</td><td></td><td>✓</td><td></td><td></td></tr></table>	One mark for each row Reject more than one tick in each row	(3)
	(b)	condom/ diaphragm/femidom/cap;		(1)
	(c)	Any four from the following • female given FSH and LH; • egg and sperms collected/extracted/removed; • egg fertilised by sperm in laboratory/ glass dish/petri dish.; • embryo formed/ develops; • embryo inserted into uterus;		(4)
	(d)	Any two from the following • passive immunity/antibodies passed from mother to baby/less chance of infection (for baby); • increases (emotional)bonding; • breast milk contains a balance/composition/combination of nutrients; • contains hormones to help development (of baby);		(2)

Total for Question 6 = 10 marks

Type MS · Describe labour and the advantages of breastfeeding

Answer 5 of 5 · Question 2 part a.ii · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a) (i)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			✓
ovary		✓	
vagina	✓		

one mark for each correct row

(3)

(ii)

- muscles/uterus wall contract;
- to widen/dilate cervix;

allow description of widening cervix

(2)

(iii)

C (a sperm cell fuses with an egg cell to form a zygote)

The following statements are incorrect because:

A haploid cells are not produced from fertilisation
B an embryo is not formed from fertilisation
D a baby is not formed at fertilisation

(1)

(iv)

luteinising hormone/LH;

(1)

Type MS · Describe labour and the advantages of breastfeeding

Answer 5 of 5 · Question 2 part a.ii · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

(b) (i)		One mark for each diagram	(2)
(ii)	mitosis → two cells, each with 23 chromosomes mitosis → four cells, each with 46 chromosomes meiosis → two cells, each with 46 chromosomes meiosis → four cells, each with 23 chromosomes	reject more than one line from each type of cell division	(2)

Total for Question 2 = 11 marks